Part C — Opinion: Investigating Stale or Irrelevant Answers

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Scenario

Six months post-deployment, users report that answers are sometimes outdated or irrelevant even though the source documents are correct and up to date.

Assumption

The system follows the Part A architecture: hybrid BM25 + FAISS retrieval with monthly re-indexing. "Source documents are correct" means the raw files on disk are current; the bug therefore lives in the indexing or retrieval pipeline, not in the documents themselves.

The Three Things I Would Investigate

1. Silent failures in the monthly re-indexing job

Why first: The most common cause of stale answers is a re-indexing job that *appears* to succeed but only partially completed — the process exited cleanly but OOM-killed mid-way, or a file permission error silently skipped a subset of documents.

How specifically:

- Run `md5sum` (or `sha256sum`) on every source document and store the checksums alongside each indexed chunk's metadata at ingest time.
- Write a reconciliation script that re-computes checksums on current source files and diffs them against the stored chunk metadata. Any document whose current checksum does not match its indexed version is a stale-index candidate.
- Cross-check the ingestion job logs for non-zero exit codes, `SIGKILL` events, or per-file error lines that were swallowed by a broad `except Exception: pass`.

If mismatched documents are found, trigger a targeted re-index of only those files rather than a full rebuild.

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2. Retrieval recall regression

Why second: Even if all documents are correctly indexed, the retrieval layer may be failing to surface the right chunks — returning topically adjacent but outdated chunks ranked above the correct ones.

How specifically:

- Build a golden evaluation set: 50 queries sampled from real user logs, each manually annotated with the correct source chunk (document ID + approximate paragraph).
- Run retrieval-only (bypass the LLM) against the current indexes and measure **recall@5** — the fraction of queries where the correct chunk appears in the top-5 results.
- If recall@5 < 0.70, investigate separately for BM25 (check vocabulary coverage for new terminology added in recent document updates — BM25 is a bag-of-words model and cannot handle vocabulary

drift without re-indexing) and FAISS (verify the embedding model's cosine similarity scores for recently-updated chunks versus older chunks on the same topic).

A recall regression here points to the embedding model no longer representing the evolved corpus well, or BM25 not having been rebuilt despite claimed success.

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3. Chunk boundary fragmentation on updated documents

Why third: Documents updated monthly are often not wholesale replacements — editors change a few paragraphs. If the new content is split differently by the chunker (e.g. the policy number moved to a different sentence boundary), the answer-critical fact may now straddle a chunk boundary and appear in neither retrieved chunk in full.

How specifically:

- Sample the 20 queries users flagged as returning outdated answers. For each, retrieve the top-3 chunks and manually verify whether the complete answer exists within a single chunk or is split across adjacent chunks.
- If fragmentation is found in > 30% of sampled failures, increase chunk overlap from 10% to 25% (in the `RecursiveCharacterTextSplitter` `chunk_overlap` parameter) and rebuild the index.
- Optionally switch to a `SentenceWindowNodeParser` (LlamaIndex) approach that indexes individual sentences but retrieves a ±2-sentence window around each hit, which is more robust to boundary drift without requiring a full overlap increase.

This is investigated third because it requires manual inspection and is harder to fix systematically than the first two.